Raphaël Padiou

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Education

MPRI – Master Parisien de Recherche en Informatique (M2) *Sept. 2026 – 2027*

- Research master in theoretical computer science (ENS, Université Paris Cité, Paris-Saclay, IP Paris); final year of the Télécom Paris engineering degree.
- Courses: online algorithms and decision-making under uncertainty (regret, competitive analysis, primal-dual); Markov chains, queueing and MDPs; search heuristics and evolutionary algorithms (multiplicative weights).

Engineering Degree – Télécom Paris *Sept. 2024 – 2027*

- Institut Polytechnique de Paris; ACCQ track in applied algebra, cryptography and coding theory.
- Relevant coursework: statistics, machine learning, optimization, game theory, algorithms for algebra and arithmetic, error-correcting codes.
- Implemented cryptographic primitives in Rust throughout the year (longstanding language, used daily since early studies).

Classe Préparatoire aux Grandes Écoles – PCSI → PC* *Sept. 2022 – June 2024*

- Intensive two-year curriculum in mathematics and physics.

Research & Implementation Projects

Distributed Consensus Protocols & Byzantine Fault Tolerance *Jan. – June 2025*
Artishow project, Télécom Paris · Go, gRPC, AWS

- Compared PBFT, HotStuff and Jolteon (quorum structures, $O(n^2) \to O(n)$ communication complexity); implemented a Byzantine consensus engine (s2PAC) in Go with SMR, evaluating safety/liveness under fault injection.
- Studied and implemented 2HK, an extension introduced by the course instructor resilient to key exposure; compared robustness/performance trade-offs against the baseline protocols.

SOS-Portfolio – Global Polynomial Optimization Repository *Feb. – Oct. 2026*
Algebraic optimization · Python, CVXPY, Clarabel/SCS

- Formulated an allocation problem as a degree-4 non-convex polynomial program; implemented dense and sparse Lasserre SOS/moment relaxations (Putinar's Positivstellensatz) as SDPs, exploiting correlative sparsity to cut the largest PSD block from 1326×1326 to 66×66 at $n = 50$.
- Added an a posteriori optimality certificate and benchmarked against multistart local search on 1,760 synthetic instances, negative results included.

Experience

Private Tutor – Mathematics & Computer Science *Sept. 2025 – Present*

- Weekly one-to-one teaching from high school to undergraduate and CPGE level (analysis, linear algebra, probability, algorithms, Python).

R&D Engineering Intern – LACME *Summer 2025*

- Reverse-engineered signals and prototyped PCBs (Altium) under strict power and telecom constraints.

Technical Skills

Programming:	Python, Rust, C/C++, OCaml, Julia, Go.
Libraries & Tools:	NumPy, SciPy, JuMP, MOSEK, PICOS, Git, Linux, LaTeX.
Mathematics:	Probability, stochastic processes, convex and polynomial optimization, online learning and bandits, randomized algorithms.

Reading & Interests

- **Reading:** *Probability and Computing* (M. Mitzenmacher, E. Upfal).
- **Hobbies:** trail running and races; guitar and piano, computer-assisted composition, live performance with the Télécom Paris music association; *The Farmer Was Replaced* (programming and automation game).
- **Languages:** French (native), English (fluent, academic use), German (learning).